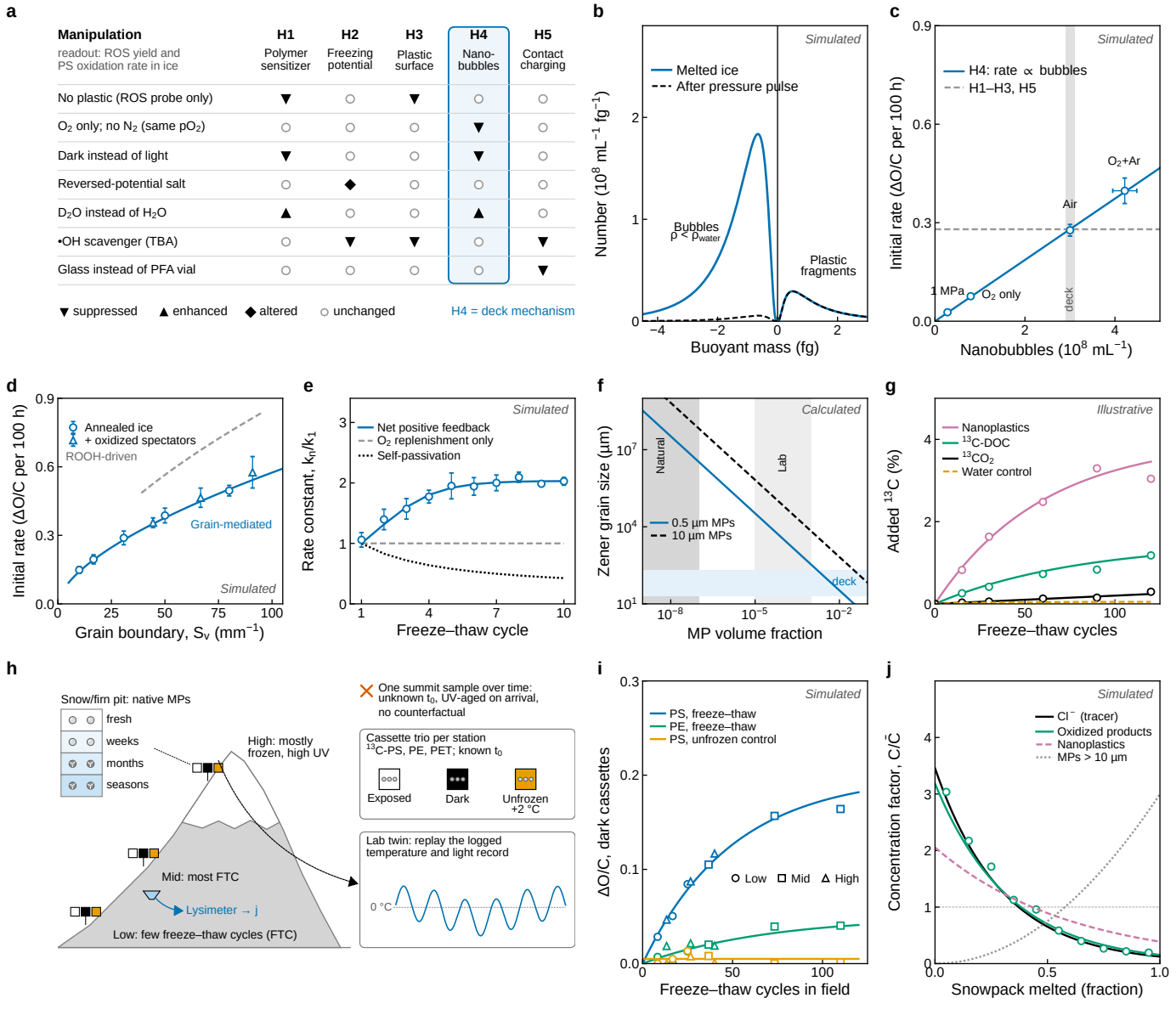

a
Manipulation
readout: ROS yield and PS oxidation rate in ice
H1 Polymer sensitizer
H2 Freezing potential
H3 Plastic surface
H4 Nano-bubbles
H5 Contact charging
No plastic (ROS probe only)
O2 only; no N2 (same pO2)
Dark instead of light
Reversed-potential salt
D2O instead of H2O
•OH scavenger (TBA)
Glass instead of PFA vial
▼ suppressed ▲ enhanced ◆ altered ○ unchanged
H4 = deck mechanism
b
Simulated
Melted ice
After pressure pulse
Bubbles ρ < ρwater
Plastic fragments
Number (10^8 mL^-1 fg^-1)
Buoyant mass (fg)
c
Simulated
H4: rate ∝ bubbles
H1–H3, H5
O2+Ar
Air
1 MPa
O2 only
deck
Initial rate (ΔO/C per 100 h)
Nanobubbles (10^8 mL^-1)
d
Annealed ice
+ oxidized spectators
ROOH-driven
Grain-mediated
Simulated
Initial rate (ΔO/C per 100 h)
Grain boundary, Sv (mm^-1)
e
Simulated
Net positive feedback
O2 replenishment only
Self-passivation
Rate constant, kn/k1
Freeze–thaw cycle
f
Calculated
Natural
Lab
0.5 µm MPs
10 µm MPs
deck
Zener grain size (µm)
MP volume fraction
g
Illustrative
Nanoplastics
13C-DOC
13CO2
Water control
Added 13C (%)
Freeze–thaw cycles
h
Snow/firn pit: native MPs
fresh
weeks
months
seasons
High: mostly frozen, high UV
Mid: most FTC
Lysimeter → j
Low: few freeze–thaw cycles (FTC)
One summit sample over time: unknown t0, UV-aged on arrival, no counterfactual
Cassette trio per station
13C-PS, PE, PET; known t0
Exposed
Dark
Unfrozen +2 °C
Lab twin: replay the logged temperature and light record
0 °C
i
Simulated
PS, freeze–thaw
PE, freeze–thaw
PS, unfrozen control
Low
Mid
High
ΔO/C, dark cassettes
Freeze–thaw cycles in field
j
Simulated
Cl− (tracer)
Oxidized products
Nanoplastics
MPs > 10 µm
Concentration factor, C/C̄
Snowpack melted (fraction)